Dynamic Planet

MIT Science Olympiad 2022

Introduction

The test is separated into 23 sections. Each section has 4 problems. Each problem is worth 2 points; partial credit will be awarded, so do your best on each problem. If all 4 problems are solved correctly (giving a total of 8 points themselves), then 2 extra points are awarded for the section, so a total of 10 points can be earned per section. Thus, there are a total of 230 possible total points.

Note: if a question does not ask you to explain your answer, you do not need to provide an explanation.

Tiebreakers: Tiebreakers are the total score of the following sections in order: 3, 1, 11, 12, and 4.

Answer: Note to graders: I sometimes add explanations to answers even though the problem did not ask the students to explain. This is only for understanding purposes: if the question did not ask the students to explain, then no explanation is necessary for full credit. I attempt to make this clear each time it comes up with either parenthesis around explanations or explicit statements like "No explanation is necessary for full credit."

1 Topographic Map Features

In this section, you will be examining various freshwater features on several topographic maps that are shown in Figures 1 and 2.

Problem 1. Examine Figure 1a. What type of geologic feature is Long Pond?
Now examine Figure 1c. Identify the main geologic feature shown in this figure.

Answer: Oxbow lake (1 point); waterfall (1 point).

Problem 2. Examine Figure 1b. Draw a longitudinal profile of the mainstem (i.e. trunk) of Dingley Creek from near Ragged Peak to the Tuolumne River. Assume this mainstem is 10000 feet long. (The contour intervals of this figure are in feet.)

Answer: insert longitudinal profile here. 0.5 points for y-axis being elevation and x-axis for being river length/horizontal distance (all or nothing. This is required for any points to be awarded on this question.)
0.5 points for starting at 10250 ft elevation.
0.5 points for ending at **below** 8500 and above 8250.
0.5 points for having a reasonable gradient.

Problem 3. Consider the figures of Cherry Lake in Figure 2. All these maps are north-oriented (i.e. North is toward the top of the image). Which direction does Cherry Creek flow? Format your answer with two directions (e.g. from East to West, from Northeast to Southwest).

Answer: From north to south.

Problem 4. Again, consider the figures of Cherry Lake in Figure 2. Infer how Cherry Lake was formed.

Answer: It is formed by the damming of Cherry Creek (at the south end).

2 Stream Drainage Systems

Problem 5. Order the following classifications of water flow in chronological order from precipitation to just before entering a stream.

1. Sheetflow
2. Gullies
3. Rills

Answer: I, III, II. All or nothing.

Problem 6. Examine Figure 3. What drainage system is shown in Figure 3a? What drainage system is shown in Figure 3b?

Answer: Figure 3a: Dendritic (1 point). Figure 3b: Radial (1 point).

Problem 7. Suppose there are two areas of land of interest: Area A and Area B. Area A exhibits hilly terrain underlain by uniform bedrock. Which drainage system does Area A most likely have, and why? Area B is underlain by various igneous rock strata of comparable resistance. Which drainage system does Area B most likely have, and why?

Answer: Area A will form a dendritic system, the rivers are molded primarily by the nature of the terrain (hilly) (1 point).
Area B will form a rectangular system, the water will flow through the joints in the rock (1 point).

Problem 8. Consider an area that exhibits two mountain valleys adjacent to each other. The middle range of mountains is lower elevation than each of the outlining mountain ranges is. How many total drainage basins are present in this area?

Answer: 3. (Each valley is one, and both put together is another.)

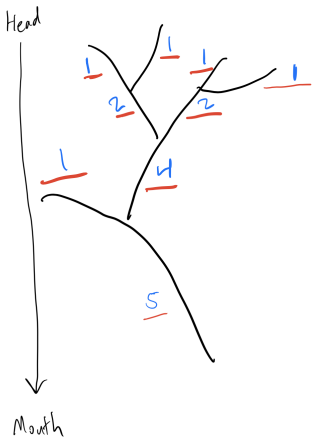


Figure 1: Section 2

3 Stream Order

Examine the stream system diagram shown in Figure 15a.

Problem 9. In Figure 15a, assume the outermost tributaries all exhibit a discharge of 1 unit. On your answer sheet, label all stream sections (including the outermost tributaries) with their respective discharge amounts.

Answer:

Problem 10. In your labeling of Figure 15a, what type of stream ordering did you exhibit?

Answer: Shreve stream order.

Problem 11. How does Horton's stream order fundamentally differ from classical stream ordering?

Answer: Horton's stream ordering is top down (outermost tributaries start as 1), while classical stream ordering is bottom up (mouth of the river starts at 1). All or nothing.

Problem 12. Consider a stream system labeled with Strahler stream order. If a certain stream section has Strahler number 4, at least how many outermost tributaries must the stream system contain? At most how many outermost tributaries must the stream system contain?

Answer: There must be at least 8 outermost tributaries: these outermost tributaries have order 1. Supposing they combine pairwise, the resultant 4 branches have order 2. Continuing in this manner, we see that there could be next 2 branches each with order 3, and finally 1 branch with order 4 (1.5 points, no explanation needed).
However, there is no upper bound: you could have an unbounded number of order 1 branches merging into a single order 2 branch (0.5 points).

4 Channel Type and Meanders

Problem 13. Suppose a certain channel has potholes. Is this channel more likely a:

A. bedrock channel

B. alluvial channel

Give your answer as one of the possible letters (A or B).

Answer: Bedrock channel.

Problem 14. Suppose a certain stream's load is mainly sand and gravel. Would you expect this stream to be a

A. braided channel

B. meandering channel

Give your answer as one of the possible letters (A or B).

Answer: Braided channel.

Problem 15. Examine Figure 5a. What is the sinuosity of the river on the left? What is the sinuosity of the river on the right? Report your answers to 2 decimal places. Would the river on the right be classified as a meandering channel?

Answer: Left: 1 (0.5 points). Right: $\frac{2 \cdot \pi \cdot 0.25}{1} = 1.57$ (1 point). No partial credit for either. Yes, the river on the right is a meandering channel since $1.57 > 1.5$. (0.5 points).

Problem 16. Consider a meandering river that, overall, runs from north to south. In one of its eastbound meanders, we define the *meander center* as the circumcenter of the triangle defined by the point where the meander begins, the farthest east point of the meander, and the point where the meander ends. (If you are unfamiliar with triangles' circumcenters, just think of it as approximately the center of a triangle.) Over time, in what direction will the meander center move?

Answer: The circumcenter will move southeast. It's critical to note that meanders migrate downward (south, in this case) as well as laterally (east, in this case). No explanation needed for full credit.

5 Capacity and Competence

Problem 17. Suppose a stream's total load is 100 units. Give as exact of a bound as possible on the stream's capacity.

Answer: Greater than or equal to 100 units. All or nothing (specifying "units" is unnecessary).

Problem 18. Consider two groups of sediment: Group A consists of sediment of grain size 0.01 mm, and Group B consists of sediment of grain size 0.1 mm. Which group of sediment requires a higher competence stream to be transported?

A. Group A

B. Group B

Give your answer as one of the possible letters (A or B).

Answer: Group B. (Requires knowing the Hjulström curve.)

Problem 19. In these next two problems, we partially derive the definition of stream power. Suppose over a certain section of a stream, the cross section area is the same, and the velocity is the same. However, the stream is still moving downhill, so it is losing potential energy. How is this potential energy used by the stream?

Answer: Friction and work against the stream perimeter (beds and banks). Either friction or work will be accepted as long as the stream perimeter is mentioned.

Problem 20. If a ball of mass m falls a distance of height h, it loses exactly mgh potential energy (where g is gravitational acceleration). Using this, write an equation that describes the power of a stream in terms of the mass of a section of water m, the gravitational acceleration constant g, the velocity of the stream v, and the slope of the stream θ.

Answer: $P = mgv \sin\theta$. All or nothing. (Note that power is energy over time.)

6 Weathering and Erosion

Problem 21. We can characterize a stream as having three general parts: a zone of erosion, a zone of deposition, and a zone of transportation. Order these zones from the head of the stream to the mouth.

Answer: 1. zone of erosion 2. zone of transportation 3. zone of deposition. All or nothing.

Problem 22. As sediment travels inside a stream, it generally takes on a rounded shape. Explain why this is the case.

Answer: Non rounded parts are more likely to strike other objects inside the stream and hence be rounded out. In other words, the sediment tends to minimize its surface area/volume ratio. Either of these two explanations will be awarded full credit.

Problem 23. Suppose there exists a river that has no bed sediment load and no suspended sediment load. Now, suppose you introduce a large amount of equally-sized sediment particles at the head of the stream and wait for an extended period of time. You then sample the most-recently deposited sediment along different sections of the river. How would you expect the sediment sampled near the mouth of the river would compare with the sediment sampled near the head of the river?

Answer: The sediment at the mouth would be smaller and more rounded. Either characteristic will be accepted for full credit.

Problem 24. Suppose you remove all the suspended and bed load sediment from a certain river. Immediately after this removal, how would you expect the amount of quarrying performed by the stream to change? How would the amount of quarrying performed change after a considerably long time has passed?

Answer: Quarrying will remain unchanged: it is mainly driven by the force of the water in the stream (which is directly related to stream velocity) (1 point).
Quarrying will still remain unchanged (the gradient/base level of the stream was not changed) (1 point).
No explanation is needed for full credit.

7 Base Level

Problem 25. Can the ultimate base level of a stream ever be lowered? Give an example if so.

Answer: Yes. Sea level is lowered (due to an increase in ice sheet volume, for example). All or nothing – must answer "yes" and give a reasonable example (virtually all examples must involve sea level being lowered).

Problem 26. Are the ultimate base levels of all streams the same?

Answer: Technically this depends on definition, so yes or no will be accepted (originally, yes was the intended answer).

Problem 27. Suppose the local base level of a stream is lowered over time. Would you expect to see the stream's floodplain exhibit terraces? What if the local base level was raised over time?

Answer: Lowered: yes (1 point). Raised: no (1 point). Terraces are fundamentally erosional features: raising base levels would result in more deposition, not erosion.
No explanation needed for full credit.

Problem 28. Can the local base level of a stream change without a tectonic event? Give an example if so.

Answer: Yes–for example, a drastic increase in precipitation. All or nothing (given example must be reasonable).

8 River Valley Landforms

Problem 29. You are analyzing the sediment composition of the banks of two rivers: River A and River B. The bank of River A is composed almost entirely of clay, while the bank of River B is composed almost entirely of sand. Assuming River A and B are located in similar climates and at similar elevations, which river would you expect to be narrower?

A. River A

B. River B

Give your answer as one of the possible letters (A or B).

Answer: River A (rivers with clay banks tend to be narrower than those with sand banks).

Problem 30. Examine the cross section of a river shown in Figure 6a. Explain why natural levees naturally take on such a shape.

Answer: As water overflows a river, velocity decreases rapidly (1 point), causing capacity to fail (1 point), leading to a tapered (triangle) shape, rather than a more rectangular shape.
Must mention both velocity decreasing and capacity falling for full credit.

Problem 31. Consider two streams that are close by: Stream A and Stream B. Suppose Stream A is experiencing headward erosion, and stream capture between the two streams occurs. Which stream was present at higher elevation before the capture?

A. Stream A

B. Stream B

Give your answer as one of the possible letters (A or B).

Answer: Stream B.

Problem 32. Although both deltas and alluvial fans are primarily depositional landforms, they depend on erosional processes to become their characteristic shapes. Explain why this is the case.

Answer: The formation of both deltas and alluvial fans require the presence of distributaries: Sediment buildup forces alternate flows from the main stream, creating distributaries, which erode new paths. These distributaries then deposit sediment of their own, creating the expanding (generally fan-like) shape of deltas and alluvial fans.
1 point for mentioning formation of distributaries, 1 point for mentioning distributaries depositing sediment themselves.

9 Velocity and Discharge I

Problem 33. Consider a specific point in a river. Can the river discharge at this point change over time?

A. Yes

B. No

Give your answer as one of the possible letters (A or B).

Answer: Yes. Due to a variable amount of water influx (e.g. from precipitation). No explanation needed for full credit.

Problem 34. Consider a meandering river. Why is its stream velocity greatest at the outside edges of its meanders? What geologic feature does this result in at the outside edge?

Answer: Over a given length and supposing no other influx of water (e.g. from a tributary), a stream must exhibit virtually the same discharge. Water that travels along the outer edge of a bend must therefore move linearly faster to compensate for the greater distance traveled. This results in a cut bank. 1 point for mentioning approximately same discharge, 0.5 points for mentioning compensation in some fashion, and 0.5 points for mentioning cut banks.

Problem 35. Suppose a river had a rectangular cross-section at some point along its path. Describe which area of this cross section experiences the greatest velocity assuming the river perimeter is uniform.

Answer: Just below the center of the river. All or nothing.

Problem 36. Consider the two cross sections of the same river at different points at time (Time A and Time B) shown in Figures 7a and 7b. At which time does the river have higher discharge at this cross section (or are they the same at both times)? At which time does the river have higher average velocity at this cross section (or are they the same at both times)?

Answer: At time B, the river has higher discharge (1 point) and higher average velocity (1 point). First, it has higher velocity since there is more water to provide more pressure (due to gravity), leading to a higher velocity. As such, the river naturally has a higher discharge at time B since its velocity and cross-sectional area is greater. No explanation needed for full credit.

10 Velocity and Discharge II

In this section, we examine the cross section diagrams shown in Figure 8. Pay special attention to the captions of each subfigure.

Problem 37. Calculate the cross sectional area for each river in the order 8a, 8b, and 8c. Report your answer to two decimal places.

Answer: 42 m^2 for 8a, 8b, 21 m^2 for 8c. 1 point for getting one correct, 1.5 for two, and 2 for all three. Units not necessary.

Problem 38. Now calculate the wetted perimeter for each river in the order 8a, 8b, and 8c. Report your answer to two decimal places.

Answer: Involves Pythagorean theorem.
Rectangle: 20 m. Trapezoid: $8 + 2\sqrt{37} = 20.17$ m. Triangle: $2\sqrt{58} = 15.23m$. 1 point for getting one correct, 1.5 for two, and 2 for all three. Units not necessary. Note that the perimeter exposed to air is not included in the wetted perimeter.

Problem 39. Given this, rank the rivers (8a, 8b, and 8c) from least discharge to greatest discharge.

Answer: Using the Chezy equation, the cross-sectional area dominates the discharge value, so 8c has the least discharge. Since the remaining two have the same cross-sectional area, we can just compare wetted perimeter. So the order is 8c, 8b, 8a. All or nothing.

Problem 40. Which of the three rivers 8a, 8b, and 8c is least likely to exist in nature?

Answer: The trapezoidal river. (Undercutting would lead to bank collapse.)

11 Springs and Geysers

Problem 41. Explain the process by which a geyser erupts.

Answer: Water near the surface of a geyser chamber slowly evaporates off (0.5 points), lessening the pressure on the rest of the water in the chamber (0.5 points). This lowers the boiling point for deeper water (0.5 points), allowing that water to boil and evaporate (0.5 points).
This essentially causes a positive feedback loop that results in a dramatic decrease in water pressure and hence eruption of water from the geyser.

Problem 42. Figure 9 shows three graphs showing temperature of water versus depth in a geyser chamber. Which graph is most realistic?

A. Graph 1

B. Graph 2

C. Graph 3

Give your answer as one of the possible letters (A, B, or C).

Answer: Graph 3 is most realistic. Water is hotter than its boiling point at the bottom of the geyser. No explanation needed for full credit.

Problem 43. Suppose the ground is tangent to the water table at some point. (A line is tangent to a curve if the line and curve intersect at a point but do not cross each other). Would you expect a spring to form at this point?

Answer: No. (The water table must intersect the ground.)

Problem 44. Suppose a geologic chamber causes a geyser chamber to expand drastically in width. After precipitation refills the chamber, would you expect future geyser eruptions to be

A. more powerful

B. less powerful

C. the same as geyser eruptions before?

Give your answer as one of the possible letters (A, B, or C).

Answer: Less powerful. A wider chamber would allow for more convection of water, making the temperature gradient less steep, thereby making the geyser less powerful. No explanation is needed for full credit.

12 Groundwater Movement and the Cone of Depression

Problem 45. If the water table is above the surface of a stream, the stream is a *gaining stream*. If the water table is below the base of a stream, the stream is a *losing stream*. If the water table is between the surface and the base of a stream, is the stream a

A. *gaining stream*

B. *losing stream*?

Give your answer as one of the possible letters (A or B).

Answer: Losing stream. The area of the stream between the water table and the surface of the stream would lose water to the ground. No explanation needed for full credit.

Problem 46. The cone of depression is a feature that forms around the bottom of a well that is pumped heavily: it is a cone-shaped zone of groundwater-free ground. In the next few problems, we will discuss why this zone is cone-shaped in particular.

Suppose a well was pumping an extremely thin aquifer that is underlain by impermeable bedrock. For the purposes of this exercise, we can just think of this aquifer as a flat plane. What shape would the well's depression take in this case, and why?

Answer: The negative pressure introduced by the well will act equally on equidistant groundwater from the center of the bottom of the well (1 point, must mention equal force on equidistant points). So the shape will be a circle (1 point). Must have the first point to get the second point.

Problem 47. Now suppose the aquifer does actually have depth. Which force is now present in this hypothetical example that wasn't important in the previous, flat-aquifer example?

Answer: Gravity.

Problem 48. With this additional force in mind, explain why the depression takes on a cone shape.

Answer: The negative pressure from the well combined with gravity create a net force that points diagonally downward toward the well.

13 Saltwater Intrusion and Groundwater Chemistry

Problem 49. In saltwater intrusion, does the saltwater primarily penetrate

A. laterally into an aquifer

B. vertically into an aquifer?

Give your answer as one of the possible letters (A or B).

Answer: Primarily vertically. (Vertical wells create an upward pressure.)

Problem 50. Explain why aquifers form a lens shape near ocean coastlines.

Answer: Saltwater from the ocean coastline enters the ground underneath the land surface in a triangle formation (1 point for mentioning saltwater entering) since salt water is more dense than freshwater (1 point).

Problem 51. Explain how carbonate forms in groundwater from dissolution of CO_2 gas by giving the necessary chemical equations.

Answer:

$$CO_2 + H_2O \longleftrightarrow H_2CO_3 \longleftrightarrow HCO_3^- + H^+ \longleftrightarrow CO_3^{2-} + 2\,H^+.$$

1.5 points for the equations set up correctly, 0.5 points for using double arrows. No states necessary.

Problem 52. Recall Le Châtelier's principle from chemistry: if equilibrium of a reaction is disturbed, the system adjusts to attempt to return to equilibrium. So if an immense amount of reactants are added, the chemical reaction will form the products at a faster rate. Using Le Châtelier's principle, explain why the removal of CO2 from groundwater induces calcite precipitation. You must give the relevant chemical equation.

Answer: In the dissolution of calcite, given by the equation

$$CaCO_3 + 2\,H^+ \longleftrightarrow H_2CO_3 + Ca^{2+},$$

more reactants than products are removed.
The equation could be reversed, but the logic must be the same.

14 Karst Topography

Problem 53. What is the name of the landmark shown in Figure 10a?

Answer: Krs Plateau, also known as the Karst Plateau. Either will be accepted for full credit.

Problem 54. Figure 10b is a cross-section diagram showing various caverns found in an area that were created by rivers (the freshwater features have been omitted). Which side of the diagram is closer to the surface?

A. Top

B. Bottom

Give your answer as one of the possible letters (A or B).

Answer: A. The top. The side with smaller caverns is closer to the surface. The river that corresponds to the cavern cuts into the ground overtime, lowering the water table. This downward movement eventually slows as the local base level is reached. As such, the caverns further down are closer to the water table for longest, so they become the largest caverns. No explanation is needed for full credit.

Problem 55. Figure 10c shows the cross section of a speleothem. Give a more precise name for this speleothem.

Answer: Stalactite. (The cross section is hollow.)

Problem 56. Using the associated letters (A, B, C, and D), rank the following in biomes from least likely to have extensive karst topography to most likely.

A. Precipitation: 80 in/year. Extensive flora.

B. Precipitation: 80 in/year. Sparse flora.

C. Precipitation: 40 in/year. Sparse flora.

D. Precipitation: 20 in/year. Extensive flora.

Answer: D, C, B, A. All or nothing. Precipitation is most important, followed by extent of flora (which, all else being equal, indicates the relative amount of carbon dioxide trapped in the ground due to the decaying of plants)

15 Lake Types I

In this section, we will explore Hutchinson's lake types as described in *A Treatise on Limnology*.

Problem 57. Which of Hutchinson's lake types (described in A Treatise on Limnology) is most abundant?

Answer: Glacial.

Problem 58. Which of Hutchinson's types is Lake Tahoe (found on the border of California and Nevada)?

Answer: Tectonic.

Problem 59. Which famous U.S. lake is pictured in Figure 11a?

Answer: Crater Lake.

Problem 60. Explain the formation of an aeolian lake.

Answer: Wind-blown sediment (usually sand) creates an enclosed area where water can pool and form a lake. 1 point for mentioning wind-blown sediment, 1 point for mentioning that it forms an enclosed area. Must have the first point to get the second point.

16 Lake Types II

You are examining two lakes (Lake A and Lake B) that experience very similar climates and are located at the same latitude. Initially, Lake A has two rivers flowing into it and one river flowing out. Lake B has one river flowing into it and no rivers flowing out. All rivers considered are of similar magnitude and characteristics.

Problem 61. Which lake is more likely to be a saline lake?

A. Lake A

B. Lake B

C. Both are equally likely

Give your answer as one of the possible letters (A, B, or C).

Answer: B. Lake B is more likely to be a saline lake. Lake A is open, and Lake B is closed. It is therefore much more likely Lake B is saline: salts accumulate but cannot leave (in substantial amounts) a closed lake. Evaporation then causes the salinity to increase. No explanation needed for full credit.

Problem 62. Now suppose a long time passes, and Lake B now has had one river flowing out for a substantial amount of time. Lake A remains unchanged. Now, which lake is more likely to be a saline lake?

A. Lake A

B. Lake B

C. Both are equally likely

Give your answer as one of the possible letters (A, B, or C).

Answer: C. Both are equally likely. (Inflow matches the outflow in both lakes, so salts likely would not accumulate.)

Problem 63. What geologic phenomenon or event could have caused the increase in precipitation which led to Lake B having an outflowing river?

Answer: Glacial melting. In general, some event that causes an increase in the available liquid water (locally or globally).

Problem 64. Now suppose all outflowing rivers from both lakes stop flowing, and a long time passes. After this, which lake is more likely to be a saline lake?

A. Lake A

B. Lake B

C. Both are equally likely

Give your answer as one of the possible letters (A, B, or C).

Answer: A. Lake A is more likely to be a saline lake. It has more inflow–and therefore more sources of sediment–than Lake B does.

17 Lake Characteristics

Problem 65. What type of circulation would cause rows of debris/organic matter to form on a lake surface?

Answer: Langmuir circulation.

Problem 66. Suppose you have two jars full of freshwater: Jar A and Jar B. The two jars are identical except for their contents. In Jar A, the water is all 1°C, while in Jar B, the water is somewhere between 2-4°C. Is Jar A as a whole:

A. less dense than Jar B is

B. more dense than Jar B is

C. the same density that Jar B is

D. There is insufficient information to answer this question.

Give your answer as one of the possible letters (A, B, C, or D).

Answer: A. (Jar A is less dense since density of freshwater increases from 1°C to about 4°C.)

Problem 67. Lakes can be separated into four distinct zones: the benthic zone, the littoral zone, the limnetic zone, and the profundal zone. Order these zones from most deep/interior to most shallow/exterior.

Answer: Benthic zone, profundal zone, limnetic zone, and littoral zone. All or nothing.

Problem 68. You sample the sediment from a lake bed in two places and find that one sample (Sample A) generally contains larger sediment than the other (Sample B). Your partner analyzes the two samples and finds that one (Sample X) is more oxidized than the other (Sample Y). Which sample does sample A most likely correspond with?

A. Sample X

B. Sample Y

Give your answer as one of the possible letters (A or B).

Answer: B. Sample A likely is the same as sample Y. (A has larger sediment, meaning its source is likely the interior of the lake bed, where the sediment is less oxidized.)

18 Lake Stratification

Problem 69. During which season do dimictic lakes exhibit inverse temperature stratification?

Answer: Winter.

Problem 70. Consider a stratified lake. The colder water is, the more dissolved oxygen it can contain. However, the epilimnion has a higher concentration of dissolved oxygen than the hypolimnion does. Explain why this is the case.

Answer: Epilimnion is nearer oxygen (since it is the topmost layer), so more oxygen will dissolve in it. Moreover, since the lake is stratified, there is little to no mixing between the layers, so oxygen does not flow from the epilimnion to the hypolimnion. 1 point for mentioning epilimnion is nearer oxygen, and 1 point for mentioning stratification prevents mixing.

Problem 71. Explain how a frozen amictic lake may still exhibit some amount of mixing.

Answer: Either by solar heating through the ice or heating from the sediment. Accept either reason for full credit. Any answer involving wind is incorrect.

Problem 72. Given the previous problem, can amictic lakes still exhibit internal waves? Explain why.

Answer: Yes. Amictic lakes are stratified. All or nothing. (The previous problem is more of a red herring–that mixing is on a small scale, and doesn't imply there is no stratification.)

19 Wetlands

Problem 73. Explain the difference between a marsh and a swamp.

Answer: Swamps are forested, while marshes have herbaceous vegetation. All or nothing.

Problem 74. What is peat? Explain why bogs have a relatively high amount of peat.

Answer: Peat is partially decomposed organic matter (1 point). Bogs have lots of peat because of a low amount of oxygen in their soils (1 point). Must get the first point to get the second point.

Problem 75. Examine the diagram of a certain wetland's water budget in Figure 12a. P represents precipitation, ET represents evapotranspiration, S_o represents surface output, and G_o represents ground output. V represents volume, and t represents time. What is the missing value for P?

Answer: $102 + 24 + 19 = 145$. No work necessary for full credit.

Problem 76. Given your answer to the previous question, what type of wetland is being diagrammed in 12a? Explain.

Answer: Bog, since the influx is dominated by precipitation. All or nothing.

20 Land Use

Problem 77. Which river is the Mississippi river threatening to divert into, thereby abandoning the Mississippi river delta? (Much effort has been made to prevent this diversion.)

Answer: The Atchafalaya River.

Problem 78. Suppose a river that has manmade levees at one section floods. Would the height of the flood upstream of the levees be

A. greater than the height of the flood at the levees

B. less than the height of the flood at the levees

C. equal to the height of the flood at the levees?

Give your answer as one of the possible letters (A, B, or C).

Answer: Less.

Problem 79. Again, suppose a river that has manmade levees at one section floods at that section when experiencing a discharge of x. Suppose the levees were then removed from that river, and some time later, the river experiences a discharge of x again. Would the new flood be greater than, less than, or equal to the height of the first flood?

A. greater than the height of the first flood

B. less than the height of the first flood

C. equal to the height of the first flood?

Give your answer as one of the possible letters (A, B, or C).

Answer: B. Less. (Levees increase flooding at a given discharge.)

Problem 80. Suppose a well has been over-pumped, causing land subsidence. The well is then left unused for some years, only to be over-pumped at the same rate later on. This time, the land subsidence is much more severe (i.e. happens at a faster rate). Explain why this is the case.

Answer: The first period of pumping caused compaction in what was the zone of saturation (1 point). As such, the second period of pumping requires more volume of ground to get the same volume of water (1 point), causing a faster rate of subsidence.

21 Hydrologic Cycle

Problem 81. Which U.S. state experiences the most rainfall yearly?

Answer: Hawaii.

Problem 82. Examine the table of water budget for a given area shown in Figure 13a. About much infiltration did the area experience?

Answer: $15 - 6 = 9\,\text{km}^3$. All precipitation virtually becomes either infiltration or runoff. Water in these forms are then transpired, evaporated, or carried away by moving freshwater. Neither the work nor the explanation is required for full credit.

Problem 83. The constituent elements of water (hydrogen and oxygen) can be found in different isotopes. Consider the isotope of hydrogen that contains a proton and a neutron: this isotope is known as deuterium. When water is evaporated from the ocean, would you expect the evaporated water to contain relatively more or less deuterium than the surface of the ocean?

A. More

B. Less

Give your answer as one of the possible letters (A or B).

Answer: A. More deuterium.

Problem 84. Would you expect the relative amount of deuterium to be greater in an aquifer located in New York or an aquifer located in Kansas?

A. New York

B. Kansas

Give your answer as one of the possible letters (A or B).

Answer: A. New York. (Heavier isotopes rain out first, so the coast receives more heavy isotope rain. Precipitation is the primary source of groundwater.)

22 Pollution

Problem 85. Which lake is the most radioactively polluted in the world?

Answer: Lake Karachay (also spelled as Lake Karachai or Lake Karachaj).

Problem 86. Would an aquifer formed in a region of karst topography be more or less able to purify polluted water than a standard aquifer? Or would they be equally able?

A. more able

B. less able

C. equally able

Give your answer as one of the possible letters (A, B, or C).

Answer: B. Aquifer in karst topography would be less able.

Problem 87. Examine the diagrams shown in Figure 14a that indicate sediment size and spacing of two aquifers. Which aquifer is more likely to be better at purifying contaminated groundwater?

A. left

B. right

Give your answer as one of the possible letters (A or B).

Answer: The right aquifer.

Problem 88. Suppose various plastics are polluting a lake. Would you expect a plastic with the same density as water to be flushed (i.e. removed from the lake by natural outflow) faster, the same speed as, or slower than a plastic with lower density than water?

A. faster

B. slower

C. equal speed

Give your answer as one of the possible letters (A, B, or C).

Answer: B. Slower.

23 Critical Zone Hydrology

Problem 89. What term describes an area of year-round unfrozen ground in a permafrost area?

Answer: Talik.

Problem 90. Does organic material in the critical zone increase or decrease infiltration?

A. increase

B. decrease

Give your answer as one of the possible letters (A or B).

Answer: A. Increase. (Breaks up soil.)

Problem 91. You are examining a pingo over time and find it to have collapsed in the winter months. Which of the two types of pingos can this pingo be?

Answer: Either a hydrostatic pingo (aka closed-system pingo) or a hydraulic pingo (aka open-system pingo). (Both can collapse.) Must have both for full credit–all or nothing.

Problem 92. You are examining a different pingo and find various regions of permafrost-free ground near it. Which of the two types of pingos can this pingo be?

Answer: Must be a hydraulic pingo (aka open-system pingo). (Hydrostatic pingos form in areas of continuous permafrost–at least in the immediate vicinity.)

Image Sheet



(a)

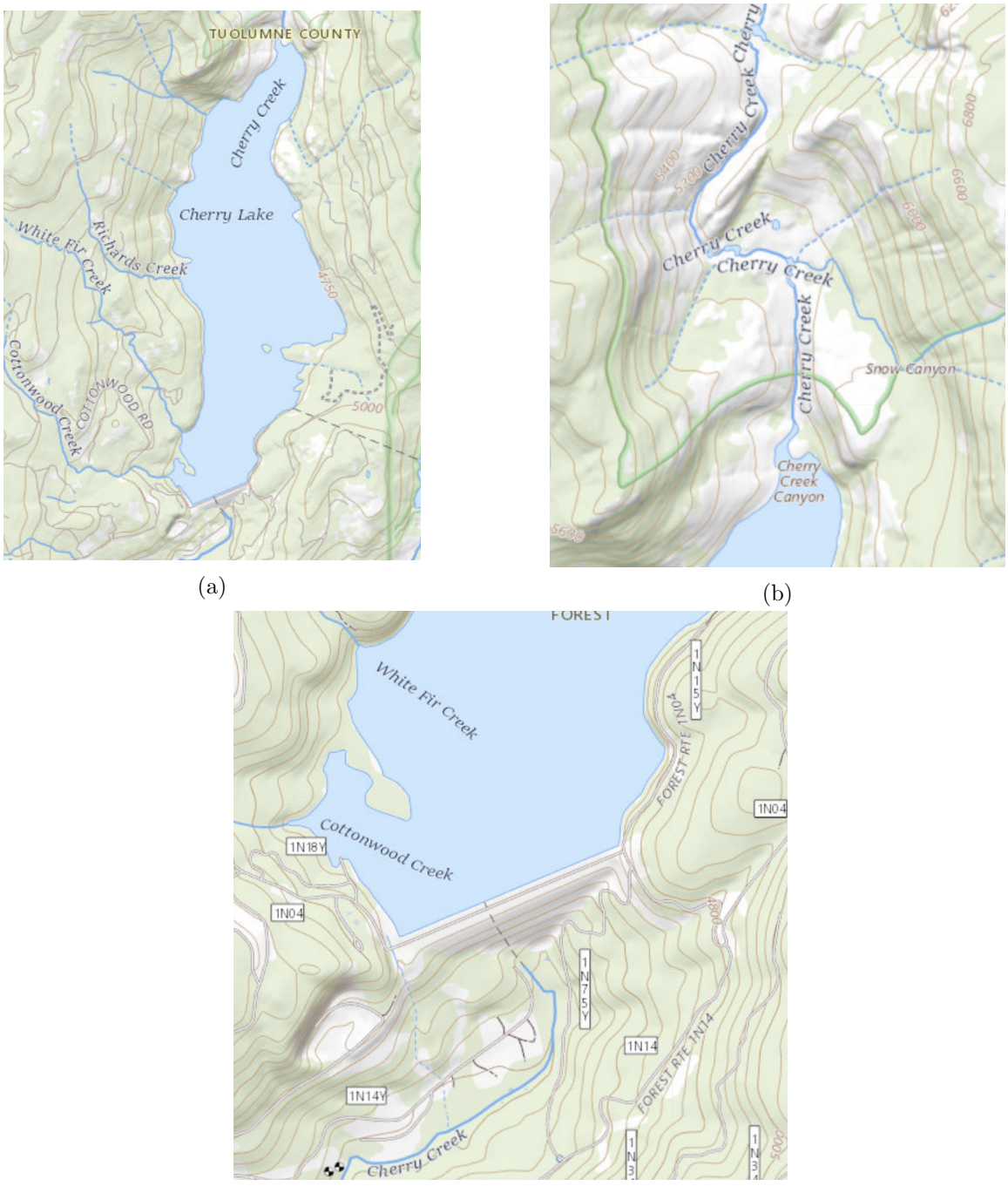


(b)



(c)

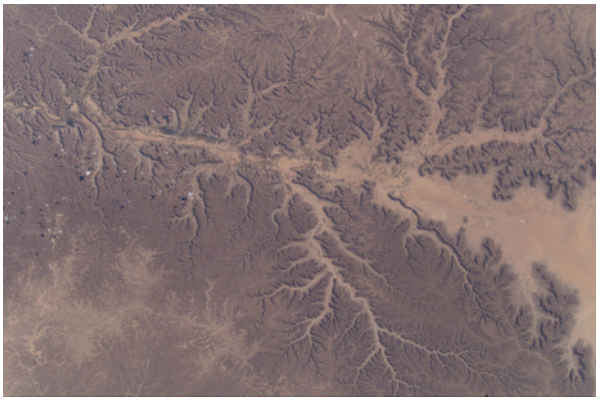
Figure 1: Section 1



(a) (b)

(c)

Figure 2: Section 1

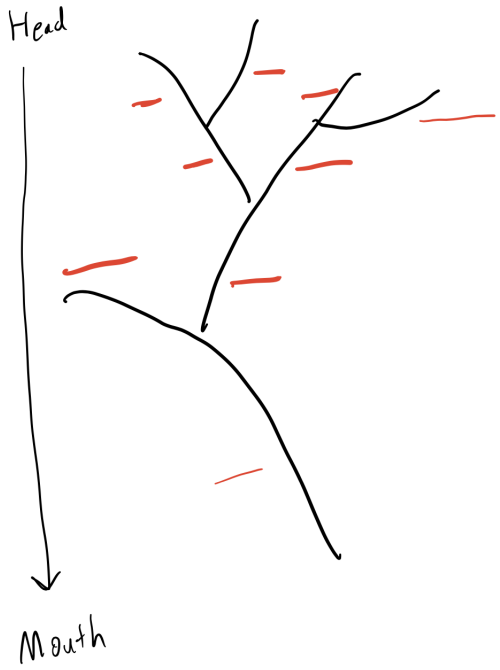


(a)



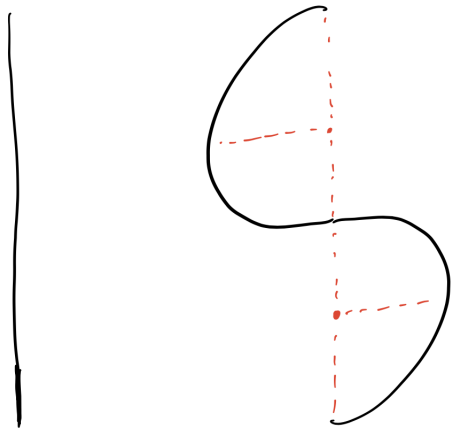
(b)

Figure 3: Section 2



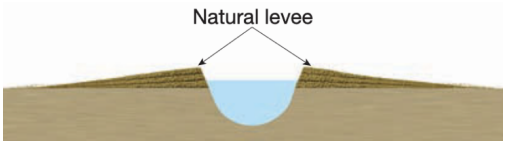
(a)

Figure 4: Section 3



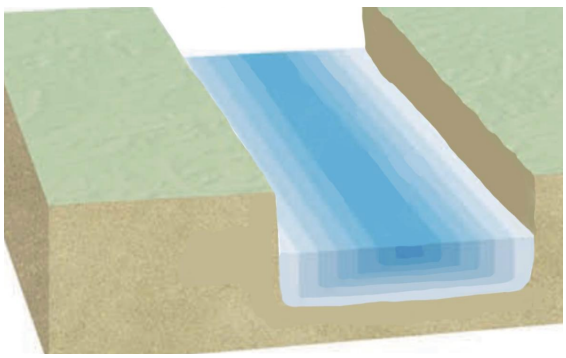
(a) Assume the straight-line distance from head to mouth for both rivers is 1km. Assume the dashed red lines in the river on the right all represent the same distance (i.e. 0.25 km). You may assume the river on the left is straight. You may also assume that the river on the right is composed of two semi-circles.

Figure 5: Section 4

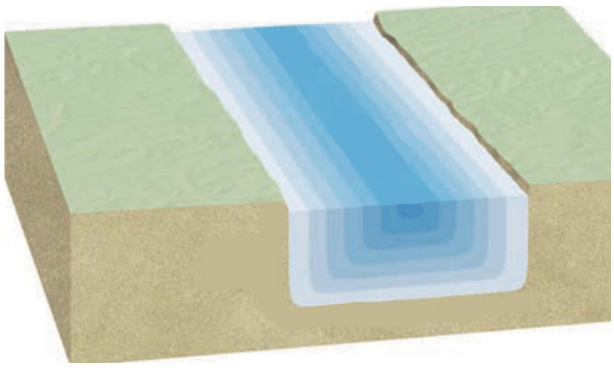


(a)

Figure 6: Section 8

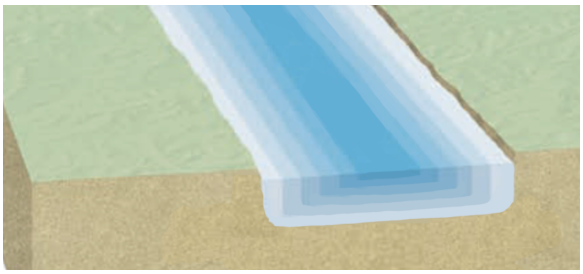


(a) River at Time A.

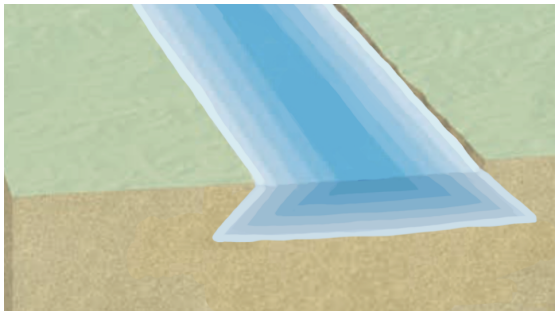


(b) River at Time B.

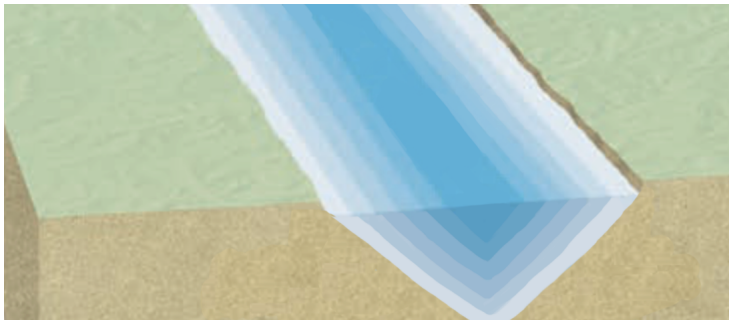
Figure 7: Section 9



(a) Rectangular Cross Section. Assume that the surface length of the cross section is 6 m and the depth is $7m$.

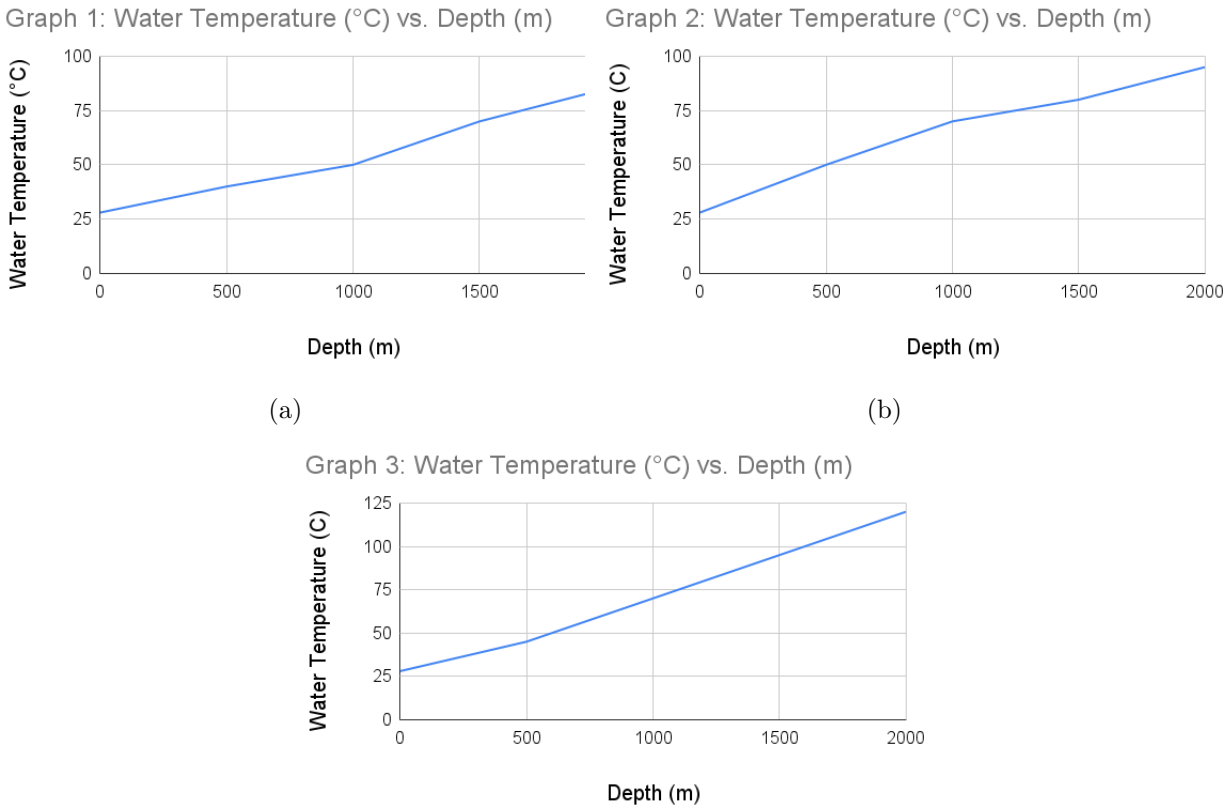


(b) Isosceles Trapezoidal Cross Section. Assume that the surface length of the cross section is $6m$, the base length of the cross section is $8m$, and the depth is $6m$.



(c) Isosceles Triangle Cross Section. Assume that the surface length is $6m$ and the depth is $7m$.

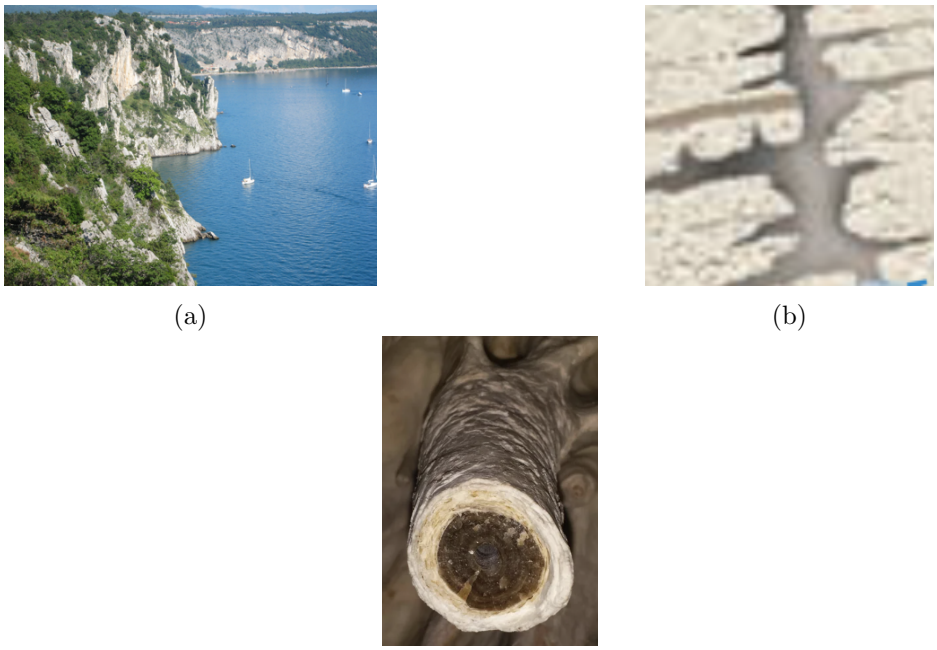
Figure 8: Section 10. Note that figures are not necessarily drawn to scale.



(a) (b)

(c)

Figure 9: Section 11



(a) (b)

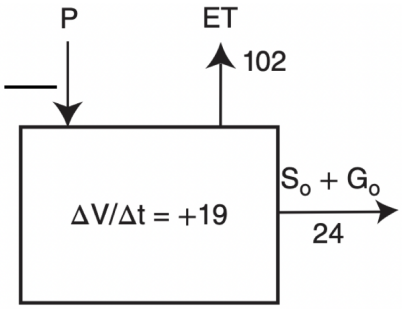
(c)

Figure 10: Section 14



(a)

Figure 11



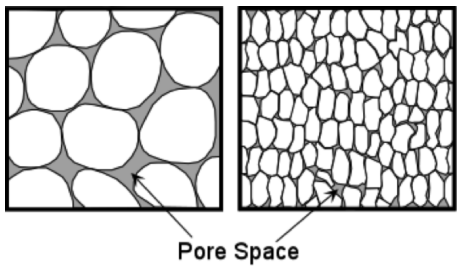
(a)

Figure 12: Section 19

Process	Amount of Water (km^3)
Infiltration	?
Runoff	6
Precipitation	15
Transpiration	1.5
Evaporation	4

(a)

Figure 13: Section 21



(a)

Figure 14: Section 22

1. _____ _____

2.

3. _____

4. _____

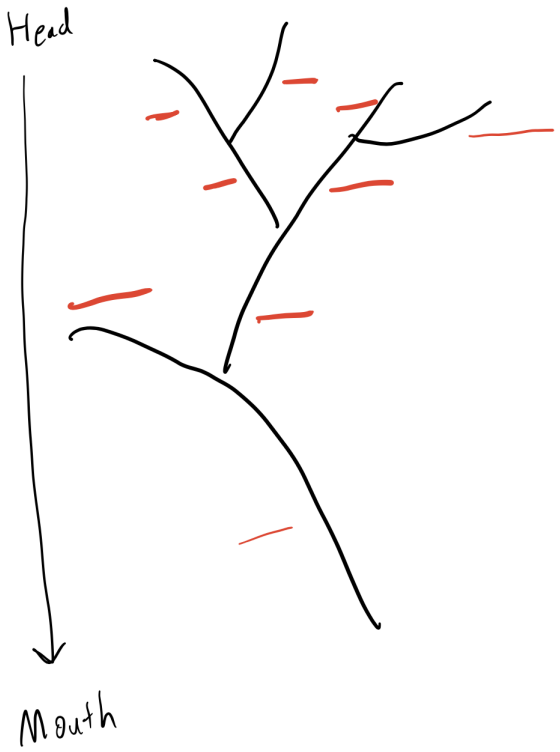
5. _____

6. _____

7. _____

8. _____

9.



(a)

Figure 15: Section 3

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